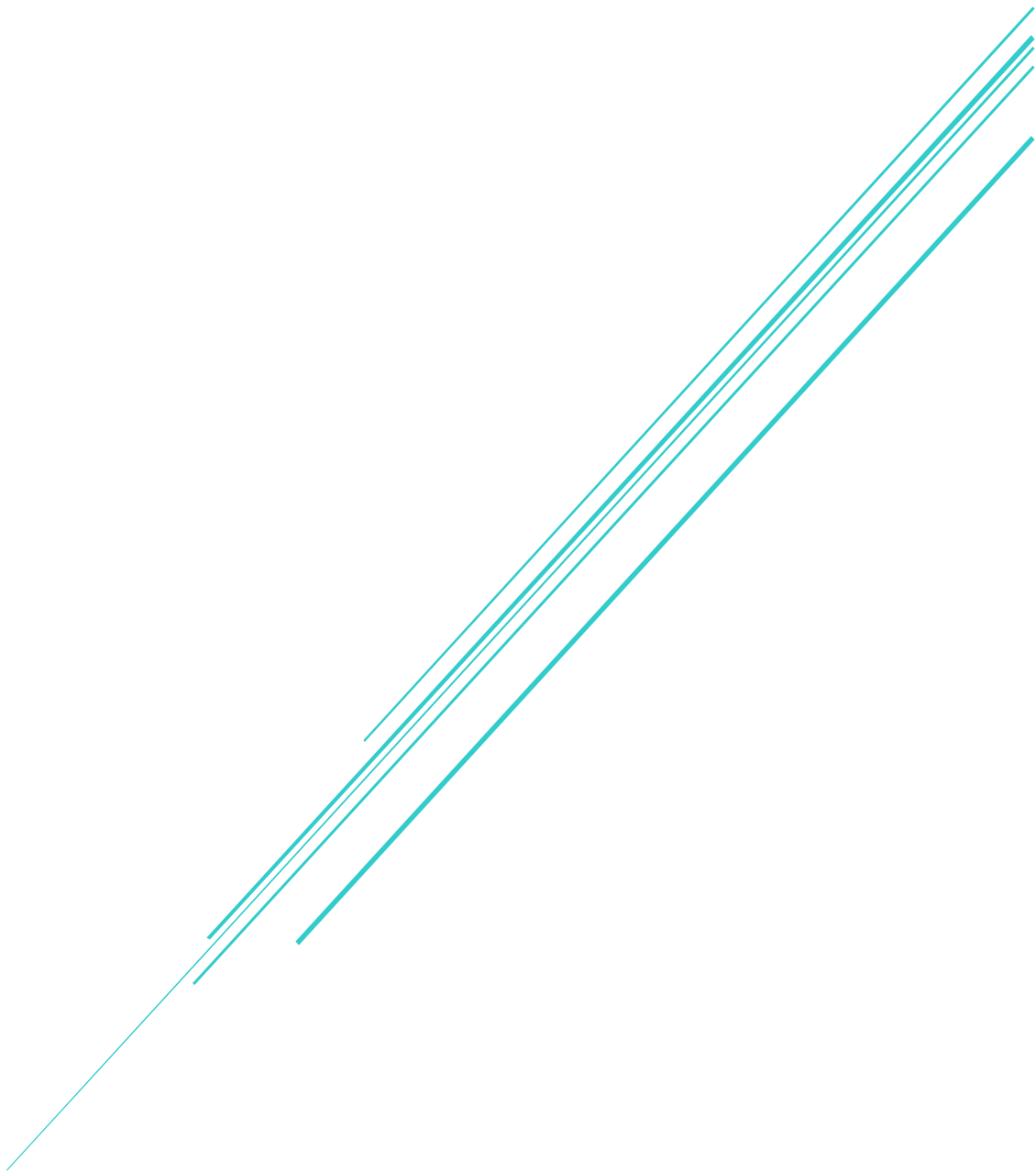


DESIGN AND IMPLEMENTATION OF AN MLOPS-ORIENTED PIPELINE FOR CMOS VOC SENSOR DATA IN EARLY VETERINARY DISEASE DETECTION-LITERATURE REVIEW

Adityanand Singh, Faezeh Kianimoravej, Sepideh Qorbani



Client
Eyuel Ayele

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1. Introduction

Gas sensing systems are widely used in applications such as environmental monitoring, industrial safety, and medical diagnostics. Despite their importance, accurately identifying gas types using sensor arrays remains a challenging task due to inherent limitations such as sensor cross-sensitivity, noise, drift, and variability across experimental conditions. These factors make it difficult to directly extract reliable and discriminative patterns from raw sensor signals.

To address these challenges, machine learning techniques have been increasingly adopted in gas sensor analysis. By transforming raw time-series signals into structured representations, machine learning models enable more effective discrimination between different gas types. However, the performance of these models is highly dependent not only on the choice of algorithm but also on the design of the data processing pipeline, including preprocessing, segmentation, feature engineering, and evaluation strategy.

A key difficulty in gas sensor time-series analysis is the presence of strong temporal dependencies within the data. Samples extracted from the same experimental run are often highly correlated, which can lead to overly optimistic performance estimates if not properly handled during data partitioning. In addition, the dynamic nature of sensor responses requires careful representation of both steady-state and transient behavior to capture meaningful gas-specific characteristics.

Given these challenges, a structured and well-designed machine learning pipeline is essential for reliable gas classification. This includes appropriate preprocessing to improve signal quality, segmentation strategies to capture temporal patterns, feature engineering to represent sensor dynamics, and evaluation methods that reflect realistic generalization performance.

The remainder of this report is organized as follows. Section 2 reviews relevant literature on preprocessing, segmentation, feature engineering, machine learning models, and evaluation strategies for gas sensor data. Subsequent sections present the implementation details, experimental results, and analysis of the proposed approach.

2. Data Preprocessing for Gas Sensor Time-Series

Data preprocessing plays a fundamental role in gas sensor data analysis, as raw sensor signals are often affected by scale differences, noise, and missing values. Effective preprocessing ensures that the data is consistent, reliable, and suitable for subsequent feature extraction and model training.

Normalization is widely recognized as a critical preprocessing step because it addresses inherent magnitude and scale differences across sensors and experimental runs (Se et al., 2025). By transforming raw resistance or voltage signals into a standardized range, normalization ensures that the data becomes uniform and comparable, facilitating downstream analysis (Se et al., 2025; An Encoder-Decoder Structure, 2023). This is particularly important in multi-sensor arrays, where individual sensors may exhibit significantly different measurement ranges. Without normalization, models may become biased toward sensors with larger numerical values, leading to distorted learning outcomes (Gutierrez-Osuna, 2002; Review on Smart Gas Sensing, 2019).

To address these issues, both global and local normalization strategies are employed. Global normalization improves comparability across sensors, while local techniques such as vector normalization compensate for sample-to-sample variations caused by analyte concentration changes and sensor drift (Gutierrez-Osuna, 2002). In addition, normalization contributes to more stable and efficient model training by reducing scale-related inconsistencies and enabling models to learn gas-related patterns more effectively (Nasri et al., 2023; Se et al., 2025). This leads to a more balanced representation of the gas “fingerprint” and improves the robustness of classification or regression outcomes (Review on Smart Gas Sensing, 2019).

Another essential preprocessing step is smoothing, which is used to address noise and measurement errors present in raw sensor signals (Review on Smart Gas Sensing, 2019). Sensor data is often affected by electrical noise and environmental fluctuations, which can obscure meaningful patterns. Smoothing techniques aim to filter out this noise while preserving the underlying signal structure required for accurate analysis (An Encoder-Decoder Structure, 2023; Saha et al., 2025). By isolating relevant signal components from irrelevant variations, smoothing enhances the quality of the data used for feature extraction (Saha et al., 2025).

Among various smoothing techniques, the Savitzky–Golay filter (SGF) is frequently employed due to its ability to reduce noise while preserving important spectral characteristics (Saha et al., 2025; Wang et al., 2023). Unlike simpler approaches, SGF maintains critical signal patterns and interaction dynamics that are essential for distinguishing between similar gas analytes (Saha et al., 2025). This preservation enables the extraction of more accurate and meaningful features, which is crucial for both supervised and unsupervised learning models. In addition to SGF, techniques such as Gaussian filtering and exponential moving averages are also used to stabilize sensor responses and ensure that extracted features reflect true gas interaction dynamics rather than transient noise (Ni et al., 2024; Zhang et al., 2025).

Handling missing values is another critical component of preprocessing, particularly in feature-engineered datasets (Nasri et al., 2023). Missing data may arise due to sensor failures or anomalies during data acquisition, and if not properly addressed, can introduce bias and degrade model

performance (Diallo & Ezin, 2023). Common strategies include imputation and data removal, both of which aim to preserve dataset integrity while ensuring consistency for model training (Diallo & Ezin, 2023).

Among imputation strategies, the use of the median is particularly common due to its robustness to outliers. Many machine learning models used in gas sensor analysis, such as LDA, SVM, and k-NN, are sensitive to extreme values, which can distort decision boundaries and negatively affect classification performance (Overcoming the Limits, 2024). Unlike the mean, the median is less influenced by anomalous readings and provides a stable representation of the central tendency of the data (Zhang et al., 2025; Overcoming the Limits, 2024).

In addition to preprocessing techniques, model selection can also influence robustness to missing data. Certain models, such as Decision Trees, are inherently capable of handling incomplete data and can still achieve strong predictive performance in the presence of missing values. Similarly, probabilistic models such as parallel neural networks can tolerate minor data gaps and noise, enhancing the reliability of gas sensing systems in real-world conditions (Overcoming the Limits, 2024).

Overall, the literature consistently highlights preprocessing as a critical factor in gas sensor data analysis. Techniques such as normalization, smoothing, and robust missing value handling play a central role in improving data quality, reducing noise, and ensuring reliable feature representation. These steps not only enhance model stability and convergence but also contribute directly to the accuracy and robustness of gas classification systems, particularly in environments characterized by noise, variability, and sensor drift.

3. Segmentation & Feature Engineering

3.1. Segmentation and Windowing Strategies in Gas Sensor Time-Series

Segmentation is a fundamental step in the analysis of gas sensor time-series data, as it transforms continuous sensor signals into structured inputs suitable for machine learning models. Among various approaches, fixed-length windowing—often implemented through sliding window segmentation or patching—has emerged as a standard strategy due to its compatibility with modern learning architectures and its ability to preserve temporal information.

Fixed-length windows are primarily adopted to satisfy the structural requirements of machine learning and deep learning models, which typically require standardized input dimensions (Se et al., 2025; Sun et al., 2023). Architectures such as Convolutional Neural Networks (CNNs), Transformers, and Long Short-Term Memory (LSTM) networks rely on consistent input shapes to perform operations such as convolution and self-attention (Ni et al., 2024; Se et al., 2025). For example, Transformer-based models employ a patching mechanism that divides raw signals into fixed-length segments, enabling the model to compute relationships between time steps more effectively (Se et al., 2025). By converting variable-length response curves into uniform sequences or matrices, fixed windowing simplifies model training and ensures that each sample contains a comparable amount of information (Ni et al., 2024; Mu et al., 2024).

This standardization also facilitates consistent feature extraction across different samples. Once signals are segmented into fixed-length windows, the same statistical descriptors or signal transformation techniques—such as FFT, DWT, or Gramian Angular Fields—can be uniformly applied, improving comparability across sensors and experimental conditions (Acharyya et al., 2022; Sun et al., 2023; Ni et al., 2024; Gutierrez-Osuna, 2002). In addition, fixed-length segmentation reduces computational complexity by focusing on informative portions of the signal while limiting redundant data (Se et al., 2025; Sun et al., 2023).

Beyond structural consistency, segmentation plays a critical role in capturing temporal dynamics. Gas sensor responses are inherently time-dependent, often characterized by transitions from baseline to steady-state behavior. Fixed-length windows enable models to analyze these temporal patterns at a fine-grained level, preserving local dependencies and interaction kinetics that are essential for distinguishing between gases with similar steady-state magnitudes (Se et al., 2025; Aly et al., 2024; Gutierrez-Osuna, 2002). This localized temporal representation is particularly important for sequence-based models, such as LSTMs, which rely on historical context to reconstruct or predict sensor responses (Aly et al., 2024).

Sliding window segmentation further enhances this capability by dividing continuous signals into overlapping or non-overlapping segments using a defined window length and sliding step (Gutierrez-Osuna, 2002; Se et al., 2025). This approach preserves the sequential structure of the data while enabling the generation of multiple samples from a single signal. As a result, models can learn detailed temporal evolution patterns that may be lost in static representations (Se et al., 2025). Additionally, this strategy supports real-time monitoring, as models can process incoming data in sequential

windows, allowing continuous analysis of streaming sensor signals (Sun et al., 2019; Overcoming the Limits, 2024).

A key advantage of sliding window approaches lies in the use of overlapping windows. By selecting a sliding step smaller than the window length, overlapping segmentation generates a larger number of training samples from the same raw data (Se et al., 2025). This is particularly beneficial in scenarios with limited data, as it increases the effective sample size and improves model learning (Se et al., 2025; Shu et al., 2023). Moreover, overlapping windows help preserve important transitional information by ensuring that critical temporal events—such as shifts from baseline to steady state—are captured across multiple segments (Se et al., 2025). This redundancy reduces the risk of information loss and improves classification robustness.

The choice of window size (or patch length) is another crucial factor that directly influences model performance. Smaller window sizes are effective for capturing local temporal features and fine-grained patterns (Se et al., 2025; Hu et al., 2026). However, excessively small windows may fail to capture sufficient contextual information, leading to reduced classification accuracy (Se et al., 2025). In contrast, larger windows provide a more comprehensive view of the signal, including broader temporal trends such as transitions from baseline to steady state (Se et al., 2025; Pareek et al., 2021). While this improves contextual understanding, it also increases computational complexity and may obscure subtle variations that are important for distinguishing similar gases (Acharyya et al., 2022; Wang et al., 2023).

Windowing strategies are also closely related to early detection capabilities. By utilizing short, fixed-length windows that focus on the initial portion of the sensor response, models can identify target gases shortly after exposure, without waiting for the signal to reach a steady state (Pareek et al., 2021; Se et al., 2025). This significantly reduces detection time, making it highly valuable in real-time and safety-critical applications. Although shorter windows may contain less information, advanced models can compensate by learning to reconstruct remaining signal dynamics from partial observations (Se et al., 2025; Aly et al., 2024).

Overall, the literature consistently identifies segmentation as a key step in structuring gas sensor time-series data for machine learning applications. Fixed-length windowing, particularly when combined with sliding and overlapping strategies, enables standardized model inputs while preserving fine-grained temporal dynamics.

Importantly, segmentation introduces a critical trade-off between temporal resolution and contextual information, where window size and overlap must be carefully selected to balance model performance, computational efficiency, and early detection capability. These findings highlight that effective segmentation is not merely a preprocessing step, but a design decision that directly influences the ability of models to capture dynamic gas response patterns and achieve reliable classification.

3.2. Feature Engineering for Gas Sensor Time-Series

Feature engineering is a fundamental stage in gas sensor data analysis, as it transforms raw time-series signals into structured representations that can be effectively utilized by machine learning models. The extraction of descriptive parameters from sensor responses plays a crucial role in pattern recognition and significantly influences the performance of classification and regression tasks (Gutierrez-Osuna, 2002). In the literature, these features are generally categorized into statistical, temporal, shape-based, and dynamic response descriptors.

Statistical and steady-state features provide a compact mathematical summary of sensor signals. Commonly extracted parameters include maximum and minimum values, mean, variance, standard deviation, median, coefficient of variation, skewness, and kurtosis (Zhang et al., 2025). In addition, steady-state features—often defined as the difference between the maximum sensor response and the baseline, or its normalized form—are widely used to represent the magnitude of the sensor response once equilibrium is reached (Wang et al., 2023; Pareek & Chaudhury, 2021). These descriptors offer a simplified yet informative representation of sensor behavior under gas exposure.

However, relying solely on static features is often insufficient, as gas sensor responses are inherently dynamic. Temporal features are therefore employed to capture how signals evolve over time. These include derivatives of resistance signals and rates of change, which describe the progression of the sensing process (Schober et al., 2021; Pareek & Chaudhury, 2021). Similarly, shape-based features focus on geometric characteristics of the response curve, such as peak values, amplitudes, and the area under the curve (AUC) (Mu et al., 2024; Zhang et al., 2025). Additional descriptors, such as slopes obtained through linear fitting of specific intervals, provide insight into diffusion–reaction dynamics and further enhance discriminative capability (Liu et al., 2025).

Dynamic response features extend this analysis by characterizing transient phases of the sensor signal, particularly during response and recovery periods. Metrics such as response time and recovery time—defined as the duration required for a 90% change in signal—are commonly used to describe sensor kinetics (Nasri et al., 2023; Feng et al., 2019). More advanced approaches involve exponential curve fitting to derive growth or decay constants, reflecting the interaction dynamics between gas molecules and sensing materials (Jang et al., 2025). In addition, techniques such as the exponential moving average (EMA) are used to estimate signal behavior during injection and cleaning phases, providing robustness against noise (Wang et al., 2023; Zhang et al., 2025).

The importance of feature engineering becomes particularly evident when considering temporal dynamics. The transient behavior of a sensor response often contains more discriminative information than steady-state values alone (Gutierrez-Osuna, 2002; Acharyya et al., 2022). Each phase of the signal—including both response and recovery—encodes essential information related to gas identity (Se et al., 2025; Ghoshal, 2020). These temporal patterns reflect interaction kinetics such as diffusion, adsorption, and reaction rates (Kwon et al., 2025; Detection and Pattern Recognition, 2025). Ignoring these dynamics can therefore reduce classification performance (Zhai et al., 2024; Ni et al., 2024).

Incorporating temporal dynamics not only improves classification accuracy but also enhances system efficiency. By leveraging early portions of the response curve, models can perform gas identification

before reaching steady state, significantly reducing detection time (Pareek et al., 2021; Se et al., 2025). This capability is particularly valuable in real-time and safety-critical applications.

First-order differences, or derivatives of the signal, play a key role in capturing short-term dynamics and transition intensity. These features quantify local changes between consecutive time steps and provide explicit information about the rate of change in the sensor response (Gas Discrimination Analysis, 2024; Acharyya et al., 2022). They are often combined with techniques such as exponential moving averages to estimate signal behavior during injection and recovery phases (An Encoder-Decoder Structure, 2023; Gas Sensor Drift Compensation, 2020). Furthermore, first-order differences are frequently used alongside raw data in neural network models, improving the ability to capture abrupt temporal variations and enhancing prediction accuracy (Gas Discrimination Analysis, 2024).

Feature engineering is particularly important for classical machine learning models, which cannot directly capture temporal dependencies in raw time-series data (Se et al., 2025; Mu et al., 2024). Algorithms such as Support Vector Machines and Random Forests treat input features as independent variables and therefore require structured representations (Gutierrez-Osuna, 2002; Pareek & Chaudhury, 2021). Engineered features address this limitation by summarizing complex sensor dynamics into a set of descriptive parameters, effectively creating an “electronic fingerprint” for each gas (Nasri et al., 2023; Feng et al., 2019).

In addition to time-domain features, signal transformation techniques are used to capture frequency-domain characteristics. Methods such as Fast Fourier Transform (FFT) and Discrete Wavelet Transform (DWT) enable extraction of spectral and multi-resolution features, revealing hidden patterns and abrupt changes in the signal (Acharyya et al., 2022; Feng et al., 2019). These transformations convert temporal information into a fixed feature space suitable for classical classifiers while preserving essential signal characteristics (Acharyya et al., 2022; Mu et al., 2024).

Finally, recent advancements have introduced deep learning approaches that perform automatic feature extraction directly from raw time-series data. Models such as 1D Convolutional Neural Networks (1D-CNN) and Long Short-Term Memory networks can learn complex spatio-temporal patterns without manual feature design (Hu et al., 2026; Sun et al., 2023). While these methods often achieve strong performance, feature engineering remains highly valuable, particularly in scenarios with limited data or when interpretability and computational efficiency are required.

Overall, the literature consistently emphasizes feature engineering as a critical component in gas sensor data analysis, enabling the transformation of complex time-series signals into informative and structured representations. A key insight is that effective feature design must capture both static characteristics and temporal dynamics, as transient signal behavior often carries the most discriminative information for gas identification.

The combination of statistical, temporal, and transformation-based features provides a comprehensive representation of sensor responses, while techniques such as first-order differences and frequency-domain analysis further enhance model sensitivity to subtle variations. Although deep learning methods offer automated feature extraction, engineered features remain highly effective and

practically relevant, particularly in settings where data availability, interpretability, and computational efficiency are important considerations.

4. Machine Learning Models for Gas Sensor Classification

Machine learning and pattern recognition methods are central to modern machine olfaction, as they enable artificial olfactory systems to overcome the inherent cross-sensitivity of gas sensors and achieve reliable analyte discrimination (Nasri et al., 2023; Mei et al., 2024). The selection of an appropriate model depends on factors such as the complexity of the sensor response, the type of extracted features, dataset size, and application requirements (Nasri et al., 2023; Acharyya et al., 2022). Broadly, the literature distinguishes between classical machine learning algorithms, which operate on engineered features, and deep learning approaches, which learn representations directly from raw sensor data (Pareek & Chaudhury, 2021; Mu et al., 2024).

Classical supervised learning models remain widely used, particularly when datasets are of moderate size and meaningful features have been extracted (Gutierrez-Osuna, 2002; Mei et al., 2024). Logistic Regression (LR) is commonly employed as a simple yet informative baseline model (Wang et al., 2025; Saha et al., 2025). Its value lies in its transparency, computational efficiency, and ability to assess linear separability of the feature space (Nasri et al., 2023; Mei et al., 2024). Strong performance from LR often indicates that the feature engineering stage has produced a discriminative representation of the gas “fingerprint” (Saha et al., 2025; Nasri et al., 2023). As such, LR serves both as a classifier and as a benchmark for evaluating the necessity of more complex models (Kroutil et al., 2022; Mei et al., 2024).

Support Vector Machines (SVM) are also widely adopted and are particularly effective for small or high-dimensional datasets (Mei et al., 2024; Feng et al., 2019; Gutierrez-Osuna, 2002). Their strength lies in constructing an optimal separating hyperplane that maximizes the decision margin (Kroutil et al., 2022; Se et al., 2025). Kernel functions allow SVM to handle nonlinear relationships by projecting data into higher-dimensional spaces (Zhang et al., 2025; Nasri et al., 2023). The Radial Basis Function (RBF) kernel is commonly used due to its flexibility in modeling nonlinear decision boundaries while balancing performance through parameters such as gamma and C (Sharma et al., 2024; Nasri et al., 2023).

Other classical models include Linear Discriminant Analysis (LDA), valued for maximizing inter-class separation and supporting dimensionality reduction, and k-Nearest Neighbors (k-NN), appreciated for its simplicity and intuitive distance-based mechanism (Gutierrez-Osuna, 2002; Nasri et al., 2023; Yang et al., 2019). Naive Bayes is sometimes used as a lightweight probabilistic alternative, although it may underperform in more complex mixture scenarios (Saha et al., 2025; Mu et al., 2024).

A major theme in the literature is the strong performance of tree-based and ensemble methods on feature-based datasets. Decision Trees (DT), Random Forest (RF), and Gradient Boosting models such as XGBoost, CatBoost, and LightGBM are consistently highlighted as effective for classification tasks involving statistical, kinetic, and spectral descriptors (Choi et al., 2026; Zhang et al., 2025; Jang et al., 2025). These models can capture nonlinear relationships and feature interactions without requiring explicit transformations (Overcoming the Limits, 2024; Choi et al., 2026).

Another advantage of tree-based models is their robustness to noise, outliers, and heterogeneous features. Sensor data is often affected by measurement errors and environmental variability, which can degrade alternative methods such as k-NN or SVM (Overcoming the Limits, 2024; Saha et al., 2025).

Random Forest, in particular, benefits from a bagging mechanism that aggregates multiple trees trained on different subsets of data, reducing overfitting and improving generalization (Diallo & Ezin, 2023; Sharma et al., 2024). Gradient Boosting extends this approach by iteratively refining weak learners to better capture residual patterns, often leading to strong predictive performance (Wang et al., 2025; Choi et al., 2026; Ma et al., 2024).

Artificial Neural Networks (ANNs) also play a significant role, especially when relationships between sensor responses and target classes are highly nonlinear (Nasri et al., 2023; Mei et al., 2024). The Multilayer Perceptron (MLP) is widely used due to its flexibility in modeling complex mappings (Gutierrez-Osuna, 2002; Nasri et al., 2023). More recent approaches include deep learning architectures such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Transformers, which can automatically extract spatiotemporal features from raw signals (Hu et al., 2026; Mu et al., 2024; Se et al., 2025; Aly et al., 2024).

Despite their representational power, neural networks are not always suitable for small tabular datasets. A consistent finding is that these models require large amounts of high-quality data and are prone to overfitting when data is limited (Nasri et al., 2023; Gutierrez-Osuna, 2002; Zhang et al., 2025). In such cases, simpler models such as SVM or Random Forest often provide more stable and reliable performance (Overcoming the Limits, 2024; Mei et al., 2024).

The literature also highlights that dimensionality reduction techniques such as Principal Component Analysis (PCA) do not always improve performance. Although PCA reduces redundancy and simplifies datasets, it is unsupervised and does not necessarily preserve class-discriminative information (Saha et al., 2025; Gutierrez-Osuna, 2002). As a result, it may prioritize variance caused by noise or drift, potentially reducing classification accuracy (Nasri et al., 2023; Mei et al., 2024). Empirical studies often report limited or no improvement from PCA, particularly when variance does not align with class separation (Aly et al., 2024; Saha et al., 2025).

Overall, the literature demonstrates that the effectiveness of machine learning models for gas sensor classification depends strongly on the relationship between feature representation, data characteristics, and model complexity. While classical models such as Logistic Regression and SVM provide reliable and interpretable baselines, ensemble methods offer strong performance by capturing nonlinear relationships and handling noisy, heterogeneous data.

A key insight is that increasing model complexity does not always lead to better performance, particularly in scenarios with limited data. In such cases, simpler models or ensemble approaches often provide more stable generalization. Although deep learning methods enable automatic feature extraction, their effectiveness is highly dependent on data availability. Consequently, model selection in gas sensing applications should be guided not by model sophistication alone, but by the balance between feature quality, dataset size, and the underlying structure of the classification problem.

5. Evaluation Strategy for Gas Sensor Classification

Evaluation strategy is a critical component of machine olfaction research, as the validity of reported performance depends not only on the selected model but also on how the data is partitioned and how performance is measured. In gas sensor time-series analysis, methodological choices such as run-level splitting, group-aware validation, and the use of multiple evaluation metrics are essential to ensure that results reflect genuine generalization rather than artifacts of the dataset structure.

A central requirement in gas sensor modeling is to split time-series data at the run level, or experiment level, rather than randomly across individual samples (Gutierrez-Osuna, 2002; Ni et al., 2024). Gas sensor signals contain strong temporal correlations, and samples extracted from the same run—such as individual time steps or overlapping windows—are often highly similar to one another (Gutierrez-Osuna, 2002). If these related samples are randomly assigned to both the training and testing sets, the resulting evaluation suffers from data leakage. In such cases, the model is effectively tested on data that closely resembles what it has already seen during training, which can produce exceedingly optimistic performance estimates (Gutierrez-Osuna, 2002). Run-level splitting addresses this problem by ensuring that all samples from a given experimental run are assigned to only one partition, thereby allowing researchers to evaluate whether the model has learned underlying chemical signatures rather than simply interpolating between near-identical observations (Ni et al., 2024; Gutierrez-Osuna, 2002).

This logic extends naturally to group-aware validation, which is particularly important when multiple samples belong to the same run, batch, or experimental session (Gutierrez-Osuna, 2002; Shu et al., 2023). Group-aware validation prevents samples from the same group from appearing in both training and validation sets, thereby reducing leakage caused by shared temporal structure, experimental artifacts, or sensor-specific patterns (Gutierrez-Osuna, 2002; Se et al., 2025). This is especially relevant in gas sensing because sensor responses are often affected by systematic sources of variance such as sensor drift, environmental changes, or batch effects (Shu et al., 2023; Pareek & Chaudhury, 2021). When group information is ignored, these drift-related patterns may appear in both training and test sets, masking the true difficulty of the problem and overstating model robustness (Shu et al., 2023). By contrast, run-level or batch-level validation forces the model to handle independent exposures and shifting measurement conditions, making the evaluation more representative of real-world deployment (Gutierrez-Osuna, 2002; Se et al., 2025).

Another important aspect of evaluation is model stability across cross-validation folds. In gas sensing applications, low variance across folds is a strong indication that the model has captured stable chemical patterns rather than relying on noise or accidental correlations tied to a particular partition of the data (Sharma et al., 2024; Zhang et al., 2025; Gutierrez-Osuna, 2002; Yang et al., 2019). Compared with a single holdout split, cross-validation provides a more robust estimate of performance because it averages results across multiple partitions, reducing the influence of an unfortunate split and ensuring that every sample contributes to both training and validation at different stages (Gutierrez-Osuna, 2002; Sharma et al., 2024). Stable cross-validation performance therefore provides stronger evidence that the model will generalize to unseen exposures in practical applications (Se et al., 2025; Zhang et al., 2025). This is particularly important for small datasets, where models—especially neural networks—may otherwise show large fluctuations in performance across folds (Acharyya et al., 2022; Zhang et al., 2025). Reporting stability measures such as standard deviation and

confidence intervals helps establish whether observed performance gains are statistically reliable and reproducible, which is especially important in high-stakes applications such as hazardous gas detection or medical diagnostics (Zhang et al., 2025; Hu et al., 2026; Wang et al., 2025).

In addition to careful data splitting and validation design, a comprehensive evaluation requires more than accuracy alone. Although accuracy provides a general measure of the proportion of correct predictions, it is often insufficient for understanding classification behavior in gas sensing tasks (Zhang et al., 2025). In imbalanced datasets, a model may achieve high accuracy simply by favoring the majority class, while performing poorly on rare but important categories (Affognon & Diallo, 2023). Moreover, accuracy does not distinguish between different types of errors, even though false positives and false negatives may have very different practical consequences in domains such as environmental monitoring or disease diagnosis (Zhang et al., 2025; Affognon & Diallo, 2023). For this reason, researchers typically report additional metrics such as precision, recall, and F1-score to obtain a more complete picture of model performance (Affognon & Diallo, 2023; Zhang et al., 2025).

Precision and recall provide complementary perspectives on classification quality. Precision measures the proportion of predicted positive cases that are actually correct, making it particularly relevant when false alarms are costly (Affognon & Diallo, 2023; Zhang et al., 2025). Recall, or sensitivity, measures the proportion of actual positive cases that the model successfully identifies, which is critical in settings where missing a target gas or disease biomarker would have serious consequences (Affognon & Diallo, 2023; Zhang et al., 2025). Because these two metrics often involve a trade-off, the F1-score is widely used as a single summary measure that balances both concerns (Affognon & Diallo, 2023; Zhang et al., 2025). As the harmonic mean of precision and recall, the F1-score only becomes high when both metrics are strong, preventing overreliance on models that perform well in one respect while failing in another (Diallo & Ezin, 2023; Zhang et al., 2025).

The F1-score is therefore considered especially important in gas sensor classification, where datasets may be imbalanced and class boundaries may overlap due to similar electronic fingerprints or environmental noise (Affognon & Diallo, 2023; Saha et al., 2025). In such cases, accuracy alone may provide a misleading picture of performance, whereas the F1-score offers a more transparent indication of whether the model is effectively capturing the underlying class structure (Zhang et al., 2025; Gutierrez-Osuna, 2002). This makes it a standard metric for comparing alternative algorithms, including Random Forests, SVMs, and deep learning models, particularly in applications where both false positives and false negatives carry significant practical costs (Zhang et al., 2025; Acharyya et al., 2022; Wang et al., 2025; Kwon et al., 2025). In many studies, the F1-score is reported alongside accuracy, AUC, confidence intervals, and statistical significance measures to support a more robust interpretation of model superiority (Zhang et al., 2025; Kwon et al., 2025).

Overall, the literature consistently shows that reliable evaluation in gas sensor modeling depends on both appropriate data partitioning and informative performance metrics. Run-level splitting and group-aware validation are essential for preventing leakage, accounting for drift, and ensuring that models are tested on truly independent data. At the same time, cross-validation stability and the use of multiple metrics—especially precision, recall, and F1-score—provide a more realistic and trustworthy basis for judging model quality. Together, these evaluation practices help distinguish

genuine generalization from artificially inflated performance and are therefore central to robust gas sensor classification.

6. Pipeline Design and Challenges in Gas Sensor Time-Series Analysis

In modern machine olfaction, the development of a time-series machine learning pipeline is a systematic process aimed at transforming raw gas sensor signals into reliable qualitative or quantitative predictions (Gutierrez-Osuna, 2002; Nasri et al., 2023). Although implementations vary across studies, the overall workflow generally follows a sequence of interconnected stages, including data preprocessing, normalization, segmentation, feature extraction, data partitioning, and model training and evaluation (Se et al., 2025; Nasri et al., 2023).

The first stage of the pipeline is data preprocessing, which is essential for improving signal quality and compensating for issues such as sensor drift, noise, and measurement variability (Gutierrez-Osuna, 2002; Review on Smart Gas Sensing, 2019). Baseline manipulation methods, including difference-based or fractional approaches, are often applied to enhance the contrast of sensor responses. Smoothing techniques such as Savitzky–Golay filtering are then used to suppress noise while preserving relevant signal characteristics (Saha et al., 2025). Following this, normalization methods such as min–max scaling or Z-score standardization are applied to reduce magnitude differences across sensors and experimental runs, improving comparability and computational stability (An Encoder-Decoder Structure, 2023; Hu et al., 2026).

Segmentation constitutes the next major stage of the pipeline. Continuous or variable-length signals are divided into fixed-length samples using strategies such as sliding windows or patching (Se et al., 2025; Pareek et al., 2021). This step standardizes model inputs while preserving local temporal dependencies that characterize the evolution of sensor responses. In addition, segmentation of early signal portions enables early detection strategies, allowing systems to identify gases before steady state is reached and reducing detection time (Pareek et al., 2021; Se et al., 2025).

Once signals have been segmented, the pipeline proceeds to feature extraction. This stage converts raw signals into descriptive parameters that capture the unique electronic fingerprint of the target gas while reducing redundancy (Gutierrez-Osuna, 2002; Review on Smart Gas Sensing, 2019). Common approaches include extracting statistical, temporal, and kinetic descriptors such as peak values, slopes, and response or recovery times (Zhang et al., 2025). Signal transformation methods such as FFT and DWT are also used to capture frequency-domain characteristics (Acharyya et al., 2022). More recent approaches employ deep learning models, such as 1D-CNNs, to perform automatic spatio-temporal feature extraction directly from raw signals (Hu et al., 2026).

A crucial step in the pipeline is data partitioning. To ensure generalization and prevent data leakage, datasets must be split at the run or experiment level so that related samples do not appear in both training and testing sets (Gutierrez-Osuna, 2002; Wang et al., 2025). Validation strategies such as k-fold cross-validation are commonly used to tune hyperparameters and assess model stability across different partitions (Acharyya et al., 2022; Kroutil et al., 2022).

The final stage consists of model training and evaluation. At this stage, models ranging from classical algorithms such as SVM and Random Forest to deep learning architectures such as LSTM are trained to map features or raw signals to target labels (Mei et al., 2024; Ni et al., 2024). Performance is

evaluated using appropriate metrics, including accuracy, precision, recall, and F1-score for classification tasks, and RMSE or R^2 for regression problems (Zhang et al., 2025; Choi et al., 2026).

Despite this structured workflow, important challenges remain, particularly in multi-stage or hierarchical systems. A key issue is error propagation, where inaccuracies in early stages negatively affect downstream components (Se et al., 2025; Schober et al., 2021). For example, if an intermediate model produces incorrect outputs, subsequent stages may fail to extract sufficient discriminative information, leading to compounded performance degradation. This problem becomes more pronounced when working with short signal windows, where available information is limited (Se et al., 2025).

To mitigate this issue, researchers often evaluate individual pipeline components separately. By training downstream models using ground-truth data rather than predicted intermediate outputs, it becomes possible to isolate the contribution of each stage and obtain a clearer assessment of its effectiveness (Se et al., 2025; Gutierrez-Osuna, 2002). More broadly, maintaining strict data partitioning and validation protocols remains essential to ensure that models learn meaningful patterns rather than propagating artifacts through the pipeline (Gutierrez-Osuna, 2002; Zhang et al., 2025).

Overall, the literature presents gas sensor analysis as an integrated pipeline in which each stage—from preprocessing to model evaluation—plays a critical role in determining overall system performance. The effectiveness of this pipeline depends not only on the quality of individual components but also on their interaction and consistency.

A key insight is that challenges such as error propagation in multi-stage systems can significantly impact reliability, particularly when intermediate outputs are used as inputs for subsequent models. Addressing these challenges requires careful pipeline design, rigorous evaluation strategies, and, where necessary, the isolation of individual components during validation. As a result, successful deployment of gas sensing systems relies on a holistic approach that balances accuracy, stability, and generalization under real-world conditions.

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